

HEREFORD, TX, USA

WMO: 720393

Lat: 34.850N

Lon: 102.333W

Elev: 1153

StdP: 88.22

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00129

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-8.9	-17.8	0.9	-1.9	-15.8	1.1	0.3	15.4	12.6	14.3	11.8	3.6	20	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	37.6	17.8	36.2	18.3	34.9	18.5	22.1	29.7	21.4	29.1	20.8	28.5	6.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.1	24.5	19.5	16.4	24.0	18.9	15.8	23.5	71.1	29.9	68.2	29.2	65.9	28.4	35.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	12.0	10.7	DB	-16.1	40.3	3.4	1.5	-18.5	41.4	-20.4	42.3	-22.3	43.2	-24.7	44.3	(4)
			WB	-16.6	24.2	3.1	4.0	-18.9	27.0	-20.7	29.3	-22.4	31.5	-24.6	34.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	3.6	5.2	10.2	14.1	18.9	25.0	25.9	25.5	21.8	15.0	8.8	4.1
	DBStd	9.33	5.25	6.05	5.30	4.66	4.69	3.12	2.52	2.73	3.38	5.04	5.21	5.20
	HDD10.0	726	206	152	61	17	2	0	0	0	0	16	82	191
	HDD18.3	2131	457	368	253	139	51	1	0	0	10	125	287	441
	CDD10.0	2509	7	18	68	139	278	448	493	481	353	170	46	8
	CDD18.3	873	0	1	2	11	68	199	234	223	113	21	0	0
	CDH23.3	11389	1	24	157	396	1144	2504	2827	2612	1302	366	55	2
	CDH26.7	5793	0	4	36	131	544	1386	1555	1430	586	117	5	0
Wind	WSAvg	4.9	4.8	5.2	5.4	5.9	5.5	5.4	4.3	4.0	4.2	4.5	4.8	4.8
Precipitation	PrecAvg	494	16	13	34	28	51	82	59	79	47	40	20	20
	PrecMax	851	45	36	110	128	146	184	127	169	161	189	91	69
	PrecMin	349	0	0	0	0	0	2	5	4	0	1	1	0
	PrecStd	115	15	12	28	28	40	49	34	38	39	44	20	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	26.3	30.1	32.6	36.7	40.3	39.5	38.0	35.3	32.2	27.4	22.7
		MCWB	9.1	11.0	12.2	12.9	15.2	16.7	18.8	18.4	17.5	15.9	11.3	9.6
	2%	DB	20.0	23.0	27.4	30.0	34.1	37.8	37.2	37.0	33.9	29.8	25.1	19.8
		MCWB	8.3	9.4	11.0	12.5	14.4	17.1	18.7	18.8	17.7	15.0	11.4	8.7
	5%	DB	17.2	20.3	24.9	27.8	32.3	36.1	35.6	35.3	32.4	27.7	22.1	17.2
		MCWB	6.7	8.3	10.6	11.9	14.4	17.7	19.2	19.4	17.6	14.6	10.2	7.6
	10%	DB	14.0	17.0	22.1	25.5	30.0	34.0	33.9	33.7	30.3	25.0	18.9	14.1
		MCWB	5.4	6.9	9.5	11.9	14.5	18.5	19.6	19.3	17.8	13.9	9.4	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	12.0	15.1	16.9	20.4	23.0	23.5	22.7	21.7	20.6	14.7	12.3
		MCDB	20.1	22.8	20.4	23.1	25.8	31.4	31.0	30.3	28.6	23.3	18.8	17.2
	2%	WB	8.7	10.4	13.4	15.6	19.0	21.6	22.2	21.9	20.7	18.9	12.9	9.9
		MCDB	17.9	20.4	21.3	21.0	25.1	29.5	29.6	29.8	27.1	22.8	19.6	17.4
	5%	WB	7.3	9.0	11.8	14.3	18.1	20.8	21.4	21.2	20.0	17.4	11.5	8.4
		MCDB	16.2	19.4	20.9	21.2	24.8	28.6	29.1	29.2	26.4	22.6	20.0	15.7
	10%	WB	5.8	7.4	10.5	13.0	17.0	20.2	20.7	20.6	19.2	16.0	10.2	6.8
		MCDB	13.4	16.6	20.2	22.6	24.0	28.2	28.8	28.6	25.6	21.4	18.4	14.1
Mean Daily Temperature Range	5% DB	MDBR	17.1	16.7	18.2	17.9	16.9	15.9	14.6	15.0	15.0	16.0	16.7	15.6
		MCDBR	20.8	22.4	22.0	21.9	20.5	18.6	16.9	16.9	17.9	19.3	21.0	20.2
		MCWBR	11.2	11.2	10.1	9.0	7.9	5.5	4.2	4.1	5.4	7.4	9.6	10.9
	5% WB	MCDBR	19.4	21.4	18.9	16.1	15.9	15.0	13.9	14.5	13.4	13.0	18.2	18.4
		MCWBR	10.6	11.2	9.7	7.8	7.2	5.7	4.6	4.4	4.6	6.2	9.1	10.5
Clear-Sky Solar Irradiance	taub	0.252	0.262	0.288	0.320	0.334	0.375	0.383	0.368	0.339	0.301	0.272	0.256	
	taud	2.592	2.578	2.507	2.413	2.437	2.363	2.384	2.395	2.450	2.541	2.550	2.577	
	Ebn at Noon	978	1000	990	964	948	905	895	905	920	938	941	951	
	Edh at Noon	75	85	100	116	115	123	120	117	105	87	77	71	
All-Sky Solar Radiation	RadAvg	3.24	4.06	5.34	6.57	7.27	7.69	7.38	6.63	5.64	4.36	3.42	2.78	
	RadStd	0.26	0.34	0.38	0.40	0.50	0.42	0.32	0.31	0.32	0.45	0.32	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)		+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96	N/A

Nomenclature: See separate page